



BACHELORS OF SCIENCE IN SOFTWARE ENGINEERING-BS(SE)

Structure of BS(SE) Program Total Credit Hours: 136 Total Courses: 44 Semesters: 8 Duration: 4 Years	Category: Courses (Credit Hours) General Education: 12 (30) Major courses: 18 (64) + 7 (21) Allied/interdisciplinary courses: 5 (15) Internship/field experience: N/A Capstone project: 2 (6) Total: 44 (136)
---	--

SCHEME OF STUDIES

S #	COURSE CODES	CLASS	COURSE TITLE	CREDIT HOURS	PRE-REQ	CO-REQ
SEMESTER 1						
1	CMC111	CORE	Programming Fundamentals	3 + 1		Programming fundamentals LAB
2	GER111	GER	Application of Information & Communication Technologies	2 + 1		App. of Info. & Comm. Tech. LAB
3	GER151	GER	Natural Science (Applied Physics)	2 + 1		Applied Physics Lab
4	GER131	GER	Calculus and Analytic Geometry	3 + 0		
5	GER121	GER	Functional English	3 + 0		
6	GER141	GER	Islamic Studies	2 + 0		
				15+3		
SEMESTER 2						
1	CMC112	CORE	Object Oriented Programming	3 + 1	CMC111	Object Oriented Programming LAB
2	GER132	GER	Discrete Structures	3 + 0	GER131	
3	CMC121	CORE	Digital Logic Design	2 + 1	GER151	Digital Logic Design LAB
4	GER142	GER	Ideology and Constitution of Pakistan	2 + 0	-	
5	GER122	GER	Expository Writing	3 + 0	GER121	
6	MTE111	MATHS	Multivariable Calculus	3 + 0	GER131	
				16+2		
SEMESTER 3						
1	CMC251	CORE	Data Structures	3 + 1	CMC112	Data Structures LAB
2	CMC222	CORE	Computer Organization & Assembly Language	3 + 1	CMC121	Computer Org. & Assembly Lang. LAB
3	SEC211	DOMAIN CORE	Introduction to Software Engineering	3 + 0	GER132	
4	CMC261	CORE	Computer Networks	3 + 1	CMC121	Computer Networks LAB
5	MTE212	MATHS	Probability & Statistics	3 + 0	MTE111	
				15+3		
SEMESTER 4						
1	CMC241	CORE	Operating Systems	3 + 1	CMC251	Operating Systems LAB

2	SEC231	DOMAIN CORE	Software Requirement Engineering	3 + 0	SEC211	
3	MTE213	MATH	Linear Algebra	3 + 0	MTE111	
4	CMC252	CORE	Analysis of Algorithms	3 + 0	CMC251	
5	MTE221	GER	Technical & Business Writing	3 + 0	GER122	
6	GERXXX	GER	Social Science I	2 + 0	MTE221	
				16+2		
SEMESTER 5						
1	SEC321	DOMAIN CORE	Software Quality Engineering	3 + 0	SEC211	
2	CMC362	CORE	Information Security	3 + 1	CMC261	Information Security LAB
3	CMC331	CORE	Database Systems	3 + 1	CMC241	Database Systems LAB
4	SEEXXX	DOMAIN ELEC	Domain Elective 1	3 + 0	-	
5	SEC311	DOMAIN CORE	Software Design & Architecture	3 + 0	SEC331	Compiler Construction LAB
				14+3		
SEMESTER 6						
1	CMC381	CORE	Artificial Intelligence	3 + 1	CMC252/ CMC371	Artificial Intelligence LAB
2	SEC312	DOMAIN CORE	Software Construction & Development	2 + 1	CMC371	Software Construction & Development LAB
3	SEEXXX	DOMAIN ELEC	Domain Elective 2	2 + 1	-	
4	ESCXXX	GER	Social Science II	3 + 0	-	
5	SEC322	DOMAIN CORE	Software Project Management	2 + 1	SEC211	Software Project Management LAB
				12+4		
SEMESTER 7						
1	CMC491	CORE	Final Year Project - I	0 + 3	CMC381	
2	SEEXXX	DOMAIN ELEC	Domain Elective 3	2 + 1	-	
3	SEEXXX	DOMAIN ELEC	Domain Elective 4	2 + 1	-	
4	SEEXXX	DOMAIN ELEC	Domain Elective 5	2 + 1	-	
5	CSC442	DOMAIN CORE	Parallel & Distributed Computing	3 + 1	CMC241/ CSC382	Parallel & Distributed Computing LAB
6	GER462	GER	Technopreneurship	2 + 0		
				11+7		
SEMESTER 8						
1	CMC492	CORE	Final Year Project - II	0 + 3	CMC491	
2	GER463	GER	Professional Practices	2 + 0		
3	GER443	GER	Civics and Community Engagement	2 + 0		
4	SEEXXX	DOMAIN ELEC	Domain Elective 6	2 + 1	-	
5	SEEXXX	DOMAIN ELEC	Domain Elective 7	2 + 1	-	
				8+5		
			TOTAL CREDIT HOURS	136		

****Campus will decide to offer a course.**